

/*SIDDIQA BAGWAN
B24CE1093

A teacher wants to analyze the performance of students in a recent examination. To understand the overall class achievement and identify trends in student performance, the teacher needs to calculate the measures of central tendency specifically, the mean, mode, and median from the student score data.*/

```
#include<iostream>
using namespace std;
int main() {
    int i,j,n;

    //Input the no.of students and their scores
    cout<<"Enter the number of students:";
    cin>>n;
    int arr[n];
    cout<<"\nEnter the scores of "<<n<<" students"<<endl;
    for(i=0;i<n;i++) {
        cout<<"Enter score of student "<<i+1<<": ";
        cin>>arr[i];
    }

    //Calculating Mean
    float sum = 0;
    for (int i=0;i<n;i++) {
        sum += arr[i]; //Sum of array elements
    }
    float mean = sum/n;
    cout<<"Mean of scores: "<<mean<<endl;

    //Median
    int temp;
    for(i=0;i<n;i++){
        for(j=0;j<n-i-1;j++){
            if(arr[j]>arr[j+1]){
                temp=arr[j];
                arr[j]=arr[j+1];
                arr[j+1]=temp;
            }
        }
    }

    cout<<"sorted in ascending order: "<<endl;
```

```

for(i=0;i<n;i++){
    cout<<" "<<arr[i];
}

float median;

if(n%2==0){
    int mid1=n/2;
    int mid2=mid1-1;
    median=(arr[mid1] +arr[mid2])/2.0;
    cout<<"\nmedian is: "<<median<<endl;

}
else{
    median=arr[n/2];
    cout<<"\nmedian is: "<<median<<endl;
}

//Mode
int maxcount=0;
int mode=arr[i];
for(i=0;i<n;i++){
    int count=0;
    for(j=0;j<n;j++){
        if(arr[j]==arr[i]){
            count++;
        }
        if(count > maxcount){
            maxcount=count;
            mode=arr[i];
        }
    }
}

cout<<"Mode is: "<<mode<<" and it occurred "<<maxcount<<" times"<<endl;

return 0;
}

```

Output

Enter the number of students:5

Enter the scores of 5 students

Enter score of student 1: 56

Enter score of student 2: 45

Enter score of student 3: 56

Enter score of student 4: 34

Enter score of student 5: 78

Mean of scores: 53.8

sorted in ascending order:

34 45 56 56 78

median is: 56

Mode is: 56 and it occurred 2 times